

Andrew Taggart

Software Engineer

(201) 417-7702 - andrew.taggart53@gmail.com - in/andrew-h-taggart/ - github.com/andrew-taggart

SKILLS

Languages: Python, R, JavaScript, HTML5, CSS3, SQL, PHP | **Tools:** Github, VSCode, Jupyter Notebook, Figma, Microsoft Office, JSON, MVC frameworks | **Databases:** MongoDB, PostgreSQL, relational database design, ER modeling, SQL querying, normalization, data extraction | **Data Science & Machine Learning:** pandas, NumPy, scikit-learn, PyTorch, SHAP, statistical analysis, exploratory data analysis, data visualization, data mining, machine learning, deep learning, classification, regression, clustering, model evaluation | **Data Engineering & Big Data:** Spark, Spark SQL, Spark DataFrames, Spark MLlib, HDFS, time series analysis | **Web & Software Development:** React.js, Vite, React Router, Leaflet, Node.js, Express.js, Django, jQuery, REST APIs, responsive UI, MVC frameworks

RESEARCH & PROJECTS

Federated Healthcare Explainability Research | Machine Learning, Federated Learning, SHAP, Permutation Importance

- Developed federated learning experiments using PyTorch and FedAvg-style aggregation to evaluate model performance and explainability across 2-client and 5-client settings for classification and survival prediction tasks.
- Built preprocessing and evaluation pipelines with pandas, NumPy, and scikit-learn, including stratified splits, one-hot encoding, standardization, Macro-F1, C-index, loss tracking, and early stopping.
- Implemented model explainability analysis using permutation importance and SHAP to compare feature-attribution stability across centralized, global federated, and local client models.

NJ-EQUIP: Spatiotemporal Analytics and Visualization Portal for Energy Equity

- Led development of an interactive energy-equity web portal that simplified environmental, utility, and demographic datasets into accessible visual tools.
- Built React frontend features including routed page views, layered geographic map interfaces, responsive UI components, and an energy burden estimator using utility costs, heating costs, and household income.
- Supported paper and website revisions with the lead and co-author; project awarded Best Paper at IEEE CCWC 2026.

Tackling Catastrophic Forgetting in Split Learning via Orthogonal Projection

- Co-authored accepted ICANN 2026 research paper; assisted with experiment setup, result review, and manuscript revisions.

Venue Purchase Portal (Ticketly) | (React.js, Python, PostgreSQL)

- Led backend development and integration with the frontend, ensuring cohesive functionality.
- Implemented and optimized the Event Browsing and Venue Discovery features, providing users with an intuitive interface to explore various events and venues, contributing to a seamless user experience.
- Collaborated closely with team members on database design and frontend development, ensuring efficient data management and a cohesive, responsive UI across all pages, including Home, Venue, Events, and Reviews.

EXPERIENCE

Equinox | Personal Trainer, New York, NY

October 2022 - December 2023

Results-driven sales and program management of luxury fitness experience

- Communicated the value of personal training, driving sales.
- Educated clients on healthy lifestyle modifications, proper technique, and safety precautions.
- Established strong relationships with clients based on trust, respect, and professionalism.

Milwaukee Tools (TTI) | Global Electronics Sourcing Analyst, Brookfield, Wisconsin

July 2021 – July 2022

Collection and analysis of data on semiconductors and relative projects

- Analyze forecasts of semiconductor cost and usage to communicate needs throughout the supply chain.
- Initiate and maintain positive relationships with semiconductor suppliers.
- Cooperate with global sourcing and procurement teams to maintain flow of supply information.
- Facilitate ongoing and new engineering projects needs for semiconductor demand.

Techtronic Industries Co. (TTI) | Field Sales and Marketing Representative, Delray Beach, Florida

July 2020 - June 2021

Managed on-site sales and marketing efforts in partnered stores.

- Maximized sell through of brands by analyzing market potential and properly merchandising products.
- Strengthened and maintained positive relationships with customers, TTI Management, and our retail partners.
- Monitored sales and analyzed monthly results to identify additional sales opportunities.
- Organized in-store product training sessions and demonstrations to educate store staff and customers.

EDUCATION

Montclair State University | M.S., Data Science | Montclair, NJ

May 2026

General Assembly | Software Engineering Immersive | Remote

April 2024

University of Miami | B.S., Human and Social Development | Coral Gables, FL

May 2020